SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**
Question Count: 5

SHORT ANSWER TEST

Code of Conduct

I S SHAMAH ALI FATHIMA(192573002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of data centers in delivering cloud services.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

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CO1 - AT 3

SHORT ANSWERS. TEST

S.SHAMAH ALI FATHIMA

192573002.

1) Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

2) Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

3) Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

4) Discuss the role of data centers in delivering cloud services.

5) Differentiate IaaS, PaaS, SaaS, XaaS with a suitable real world examples.

EFINE CLOUD COMPUTING & EX
THAT MAKE IT DIFFERENT FROM TRADITIONAL

COMPUTING:

Cloud computing is a technology that provides computing resources such as servers, storage, databases, networking & software over the internet.

Four characteristics of cloud computing.

i) On demand self service.

Users can access computing resources such as storage & servers whenever they need them without contacting the service provider.

Ex: Creating a virtual machine on AWS.

ii) Broad Network Access:

Cloud services are accessible through the internet using devices such as laptops, smartphones & tablets.

Ex: Accessing Google Drive using a mobile phone.

iii) Resource Pooling.

Cloud providers share computing resources among multiple users using a common infrastructure. Resources are allocated according to user requirement.

Conclusion:

Cloud computing offers flexibly, scalability, easy access and efficient resource utilization compared with traditional on-promises computing.

2) TRACE THE EVOLUTION OF CLOUD COMPUTING FROM HARDWARE EVOLUTION TO INTERNET SOFTWARE EVOLUTION & SHOW HOW THESE CHANGES ENABLED MODERN CLOUD PLATFORMS.

Cloud computing evolved through several technological developments in hardware, networking, virtualization & internet-based software.

Evolution of Cloud Computing.

i) Mainframe Computing.

This introduced the concept of sharing computing resources.

ii) Personal Computer Era.

Personal computers made computing available to individuals.

iii) client - server Computing.

Computing tasks were divided between clients & centralized servers.

iv) Distributed Computing.

Multiple connected computers worked together to complete tasks.

v) Virtualization Technology.

Virtualization allowed multiple virtual machines to run on a single physical server.

vi) Internet & Broadband Development.

Faster internet connections enabled users to access remote servers and application.

vii) web-Based Software & Saas.

Software began to be delivered through web browsers instead of requiring installation on every computer.

3) ROLE OF SERVER VIRTUALIZATION IN CLOUD COMPUTING.

i) Resource Sharing:

Multiple virtual machines can share the CPU, memory and storage of a physical server

ii) Cost reduction.

Organizations need fewer physical servers, reducing hardware, electricity & maintenance cost.

iii) Scalability.

Virtual machines can be created, modified or removed according to changing requirements.

iv) Isolation.

Each virtual machine operates independently, so applications running on one virtual machine are seperated from those on another.

v) Flexibility.

Different operating systems & applications can run on seperate virtual machines using the same physical hardware.

THE ROLE OF DATA CENTERS IN DELIVERING CLOUD SERVICES.

A data center is a physical facility containing servers, storage devices, networking equipment and supporting systems used to store, process and manage data.

Roles of Data Centers.

1. Data storage:

Data centers store large amounts of user data, files, databases & application information.

2. Computing and Processing:

Servers process user requests and run applications such as websites and online services.

3. Network Connectivity:

Routers and switches connect servers to the internet, allowing users to access cloud services from anywhere.

4. Virtualization:

Data centers use virtualization to run multiple virtual machines on a single physical server, improving resource utilization.

5. Security:

Security measures such as firewalls, encryption, access controls, and surveillance protect data from unauthorized access.

6. Backup & Recovery:

Data centers maintain backups and support data recovery during hardware failures or unexpected disasters.

7. High availability:

Backup power supplies, redundant servers & network connections help ensure continuous service.

Conclusion:

Data centers are the backbone of cloud computing. They provide storage, processing power, connectivity, security & reliability for delivering cloud services efficiently.

Differentiate IaaS, PaaS, SaaS, and XaaS with examples.

Feature	IaaS	PaaS	SaaS
Full Form	Infrastructure as a Service	Platform as a Services	Software as a ser
Meaning	Provides virtual, servers, storage, & networking.	Provides a Plat-form to delvelop.	Provides ready to use software over the internet
User responsibility.	Manages OS & applications.	Develops & manages application	uses the software
Main users.	IT administrators	Developers	End users.
Advantages.	Reduces hardware costs.	Simplifies development	Easy acces.
Example.	Amazon EC2.	Google App engine	Google Docs.

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